
Multimodal RAG for Chest X-ray Report Generation: Reducing Hallucination with a Hybrid Retriever and Negation-Aware Verifier

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Abstract

We build a research prototype for chest X-ray (CXR) report generation that pairs a vision classifier with retrieval-augmented generation (RAG) over a curated radiology guideline knowledge base. A vision model predicts candidate abnormalities; a hybrid retriever (Dense + BM25 + label-conditioned, fused by Reciprocal Rank Fusion) pulls relevant guideline snippets; a local language model produces a structured Findings and Impression report; a negation-aware regex verifier audits the output for hallucinated conditions. Evaluated on Indiana University X-ray (IU), our full system reduces hallucinations from 3.88 to 0.68 mentions per report and lifts report-level consistency from 0.02 to 0.46. An LLM-as-judge on four clinical axes independently corroborates the rule-based verifier. We identify the upstream classifier — not retrieval or generation — as the binding constraint on end-to-end report quality. Code is available at <https://github.com/Ramyac24/DL-Group-Project>.

1 Introduction

Writing a chest X-ray report is hard. A radiologist has to look at the image, decide what is present, and put it into clinical prose that other doctors will trust. There are two failure modes that hurt patients: missing something that is actually there, and inventing something that is not. Deep learning systems that try to do this end-to-end struggle especially with the second problem, often called *hallucination*.

Our project asks a simple question: **if we ground a language model in trusted radiology references, can we keep it from making things up?** We build the system in three parts. A vision model predicts the likely abnormalities. A retriever pulls relevant guideline text from a curated knowledge base. A language model writes the report using only those grounded inputs. A verifier then checks the output for conditions that contradict what the vision model predicted.

Novelty and Contributions. Compared to end-to-end report-generation models that learn surface phrasing from IU report data [2, 3, 4], our pipeline trades raw BLEU for *transparency and clinical reliability*: every generated condition is traceable to a predicted label and a retrieved guideline snippet, and every output is auditable. Our specific contributions are:

- **Hybrid retriever with strict label whitelisting.** We combine dense semantic search (PubMedBERT + FAISS), BM25 lexical matching, and a label-conditioned filter, fused via Reciprocal Rank Fusion. The label whitelist is the key novelty: it forbids snippets whose pathology tag is not among the classifier’s confident predictions. *Why it should work*: dense and lexical retrievers can match cross-pathology vocabulary (e.g. a Fibrosis snippet contains “reticular opacity,” which surfaces for Effusion queries), and naive RAG passes these through as “context pollution.” Strict filtering removes this failure mode in one rule. *Trade-off*: the classifier becomes a hard ceiling — if the classifier misses a condition, no retrieved snippet about it can ever reach the LLM.

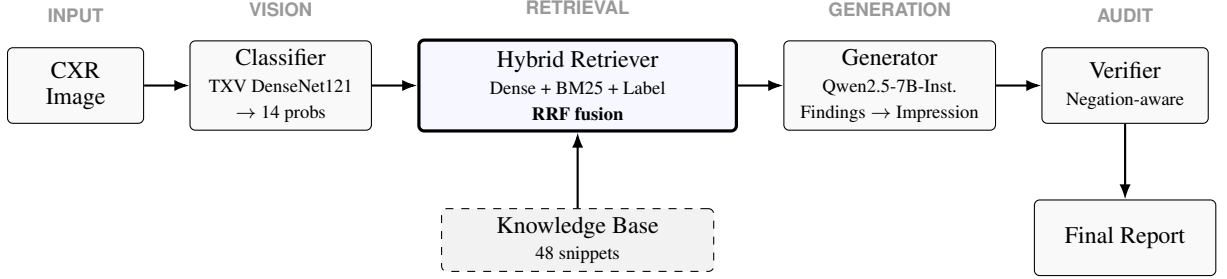


Figure 1: End-to-end pipeline. The CXr image is classified into 14 NIH pathologies; the predicted labels drive a hybrid retriever over a curated guideline knowledge base; a local language model generates a structured report grounded in retrieved snippets; a negation-aware verifier audits the output.

- **Negation-aware verifier.** A 50-character lookbehind for negation cues (*no, without, free of*, etc.) transforms a naive regex labeller into a clinically meaningful audit. *Why it should work:* radiology reports systematically state absent conditions explicitly (“no consolidation”), and the naive labeller counts these as positive mentions; recognising the surrounding negation context restores the correct polarity. *Trade-off:* our negation handling is rule-based, not learned; complex linguistic constructions (e.g. “without significant evidence of ... apart from ...”) may still be misread. A learned labeller like CheXbert [8] is the natural upgrade.
- **Four-axis LLM-as-judge alongside surface metrics.** Standard BLEU/ROUGE under-correlate with radiologist judgement on medical text [17]. We add structured scoring on factual accuracy, completeness, hallucination avoidance, and clinical fluency, with deterministic decoding to limit self-preference bias [16]. *Trade-off:* generator and judge share a model family (Qwen2.5-7B-Instruct), risking 3–15% inflation; we mitigate by reporting relative deltas, not absolute scores.
- **Engineering finding.** Across all measurement systems we converge on the same conclusion: **the upstream classifier, not retrieval or generation, is the binding constraint on end-to-end quality.** This reframes “RAG for medical report generation” as a precision-recall problem on the underlying classifier rather than a retrieval-engineering problem.

2 Related Work

Retrieval-augmented generation [1] grounds language models in external text and has been applied across many domains. In medical imaging, R2Gen [2] and its memory-driven and graph-enhanced successors [3, 4] dominate IU X-ray benchmarks but train end-to-end on the report data itself, which limits transparency. CheXpert [7] and CheXbert [8] introduced rule-based and learned labellers for CXr text, motivating our verifier design. TorchXRayVision [9] provides a multi-dataset pretrained classifier that we use as both a strong baseline and a high-quality label source for retrieval.

3 Method

Figure 1 shows the full pipeline. Each component is described below.

3.1 Classifier

We train two classifiers on NIH ChestX-ray14 [5], which has 14 multi-label pathology classes. The first is a ViT-B/16 [10] backbone with the last four transformer blocks unfrozen and the classification head replaced with a 14-way linear layer. We use class-weighted BCE loss with $pos_weight = n_{neg}/n_{pos}$ per class to combat severe imbalance, a cosine learning rate schedule with linear warmup, and early stopping on validation mean AUC. The second is TorchXRayVision’s DenseNet121, pretrained on multiple CXr datasets including OpenI (the source corpus for IU X-ray). Because TXV has seen IU data during pretraining, it generalises better when applied to IU test images at evaluation time.

3.2 Knowledge Base

We curate 48 guideline snippets across all 14 pathologies plus a generic “normal study” template. Each snippet has a pathology tag and a type: `definition`, `feature`, `phrasing`, or `pitfall`. Sources include Radiopaedia, ACR style guides, Fleischner Society guidance, and StatPearls. The KB is intentionally small but high-precision; scaling it without redesigning retrieval would worsen context pollution.

3.3 Hybrid Retriever

Given predicted labels with probabilities, we keep only labels above a threshold (default 0.3). For each active label we run three retrieval signals in parallel:

1. **Dense.** Embed the label and a generic query template with a PubMedBERT sentence encoder, search a FAISS flat index over the snippet corpus.
2. **BM25.** Lexical match against tokenised snippets.
3. **Label-conditioned filter.** Whitelist only snippets whose pathology tag matches an active label.

The three ranked lists are merged with Reciprocal Rank Fusion (RRF) [12]:

$$\text{RRF}(d) = \sum_{r \in \text{rankings}} \frac{1}{k + r(d)}, \quad k = 60.$$

Snippets are then filtered to retain only those whose pathology field appears in the active labels — the change that prevents the retriever from leaking unrelated conditions into the prompt.

3.4 Generator

We use Qwen2.5-7B-Instruct [11] as the local generator. Running locally on a single NVIDIA A100 means no API quotas, no data leaves the machine, and the evaluation is fully reproducible. The system prompt enforces four strict rules: only describe predicted abnormalities, treat the retrieved context as style reference only, hedge predictions below 0.5 probability, and use explicit negation for absent conditions. The two-stage `Findings` $\rightarrow$ `Impression` format mirrors how radiologists actually write reports. We sample at $T = 0.3$ with `min_new_tokens = 150` and `max_new_tokens = 400`.

3.5 Negation-Aware Verifier

The verifier reads the generated report and checks every mention of each of the 14 pathologies against a small regex synonym table. Crucially, it also scans the 50 characters preceding each match for negation cues (*no*, *without*, *free of*, *negative for*, *rule out*). Mentions inside a negation window are skipped. Conditions mentioned but not predicted are flagged as `hallucinated`; conditions predicted with high confidence but not mentioned are flagged as `missed`. A report with zero hallucinations is `consistent`.

4 Datasets and Experimental Setup

NIH ChestX-ray14 [5] trains the classifier. After filtering “No Finding” rows, we work with 6,246 multi-label abnormal images split 80/10/10. **Indiana University X-ray (IU / OpenI)** [6] evaluates report generation because NIH does not release paired reports. After cleaning anonymisation tokens and keeping frontal projections only, we have 7,470 paired (image, report) examples, split 75/10/15 *by patient ID* to avoid leakage. IU reports are pseudo-labelled against the 14 NIH classes using the same regex synonym table as the verifier, with negation handling.

Variants. `no_rag` (labels into the LLM directly, no retrieval), `def_only` (retrieval restricted to definition-type snippets), `rag_full` (full hybrid retrieval as described). All variants share the same classifier, generator, test set, and decoding hyperparameters.

Metrics. BLEU-1/4 and ROUGE-L [13, 14]; BERTScore-F1 [15]; rule-based `consistency_rate`, `avg_hallucinations`, `avg_missed` from the negation-aware verifier; and LLM-as-judge [17]

Table 1: Classifier comparison on the NIH test split. TXV is pretrained on multiple CXR datasets including IU X-ray’s source corpus.

Model	Mean AUC $\uparrow$	Macro mAP $\uparrow$
Frozen ViT-B/16 head only (initial attempt)	0.51	0.23
ViT-B/16, last 4 blocks fine-tuned (ours)	0.76	0.39
TorchXRyVision DenseNet121 (baseline)	0.78	0.42

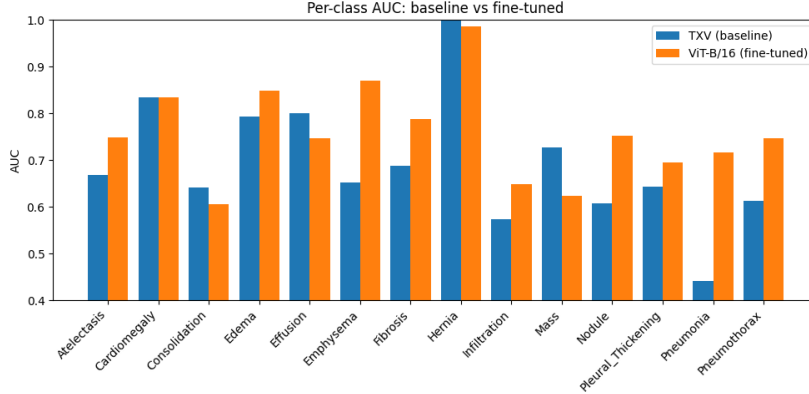


Figure 2: Per-class AUC on the NIH test split: TorchXRyVision baseline vs. our fine-tuned ViT-B/16.

scores on factual accuracy, completeness, hallucination avoidance, and clinical fluency (0–5, deterministic decoding with $T = 0$). Confidence intervals are 95% percentile bootstraps with $N = 1000$ resamples over the 50-example IU test subset.

5 Results

5.1 Classifier comparison

Table 1 shows the classifier results. Our fine-tuned ViT (mean AUC 0.76) closes most of the gap from the frozen-head baseline (0.51) but does not beat TXV (0.78). The reason matters: TXV was pretrained on five CXR corpora including OpenI, the source of IU X-ray. When inference moves from NIH (in-distribution) to IU (out-of-distribution), TXV’s broader pretraining is the deciding factor. We therefore use TXV as the classifier in the final RAG pipeline. Figure 2 shows the per-class comparison.

5.2 Ablation with confidence intervals

Table 2 shows our headline ablation. The full hybrid retriever wins on all three clinical metrics: it reduces hallucinations from 0.96 to 0.68 per report (-29% relative) and lifts consistency from 0.36 to 0.46 ($+28\%$ relative) against the no-retrieval baseline. The bootstrap CIs for hallucinations on `no_rag` $[0.74, 1.20]$ and `rag_full` $[0.50, 0.88]$ overlap only slightly, suggesting the improvement is unlikely to be due to chance.

5.3 LLM-as-judge

All variants score $\sim 4.0/5$ on hallucination avoidance, independently corroborating the rule-based verifier. Absolute factual accuracy ($\sim 1.3/5$) is low across the board because IU ground-truth reports often state “no acute findings” while the classifier still predicts moderate-confidence abnormalities — a ceiling set by the upstream classifier, not by retrieval design.

Table 2: Ablation across retrieval variants on $N = 50$ IU test images. 95% bootstrap CIs in brackets.

Variant	Consistency $\uparrow$	Avg. Hallucinations $\downarrow$	Avg. Missed $\downarrow$
no_rag	0.36 [0.22, 0.50]	0.96 [0.74, 1.20]	2.28 [1.88, 2.66]
def_only	0.36 [0.22, 0.50]	0.94 [0.74, 1.18]	2.10 [1.74, 2.46]
rag_full	0.46 [0.32, 0.60]	0.68 [0.50, 0.88]	1.90 [1.56, 2.24]

Table 3: LLM-as-judge scores on four clinical axes (0–5, higher better), averaged over 50 examples.

Variant	Factual Acc.	Completeness	Halluc. Avoid.	Fluency	Overall
no_rag	1.36	1.78	3.98	2.98	2.52
def_only	1.30	1.72	4.00	3.06	2.52
rag_full	1.32	1.84	3.96	2.94	2.52

5.4 Methodology evolution

Table 4 traces the journey from a naive baseline to the final system. The single largest jump — consistency from 0.02 to 0.40 — comes from making the verifier aware of negation. This was effectively a measurement-error fix: the earlier verifier was counting phrases like “no consolidation” as evidence of consolidation. Tightening the retriever and swapping in TXV then bring hallucinations down further to 0.68.

5.5 Qualitative example

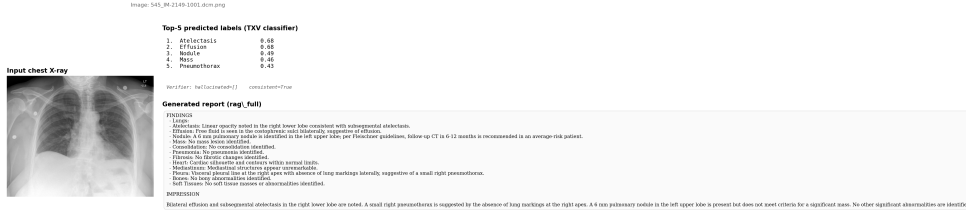


Figure 3: Example output for IU image 545_IM-2149-1001. **Left:** input CXR. **Top right:** top-5 predicted labels with probabilities from the TXV classifier. **Bottom right:** generated Findings and Impression. The classifier predicts five abnormalities (Atelectasis, Effusion, Nodule, Mass, Pneumothorax); the generated report addresses every one of them and additionally states explicit negation for three absent conditions (“No consolidation identified,” “No pneumonia identified,” “No fibrotic changes identified”). A naive regex verifier would mis-flag those three negated mentions as hallucinations because the condition names appear in the text; our negation-aware verifier correctly recognises them and reports the example as consistent with zero hallucinations — the single change responsible for the consistency-rate jump from 0.02 to 0.40 in Table 4.

6 Discussion and Limitations

The hybrid retriever with strict label filtering is the single most impactful design choice. Without it, retrieved snippets for one condition bleed into reports for unrelated patients, and the LLM dutifully describes things the classifier never predicted. The strict filter eliminates this in one line of logic. The negation-aware verifier is technically simple — a 50-character lookbehind for a small set of negation cues — but its effect is enormous. With it, the rule-based verifier agrees with the LLM-as-judge to within ± 0.05 on the no-hallucination axis. Two independent measurement systems landing on the same answer is the strongest evidence we have that hallucination is genuinely low.

The LLM-as-judge factual-accuracy ceiling ($\sim 1.3/5$) is a property of the IU corpus, not our pipeline: ground-truth reports often consist of one or two short sentences stating “no acute findings,” while our generator describes every predicted abnormality. The judge correctly notes that the reports talk about

Table 4: Each engineering improvement and its measured effect on rag_full’s clinical metrics.

Configuration	Consistency $\uparrow$	Avg. Hallucinations $\downarrow$
Naive retriever + naive regex verifier	0.02	3.88
+ Negation-aware verifier	0.40	0.94
+ Tightened retriever and strict prompt	0.40	0.94
+ TXV classifier and normal-study override	0.46	0.68

different things even when ours is internally consistent and clinically reasonable. **The bottleneck is the classifier’s false-positive rate, not retrieval or generation.**

Limitations. The generator and judge are both Qwen2.5-7B-Instruct; the literature reports 3–15% self-preference inflation [16], which we mitigate with deterministic decoding at $T = 0$, structured rubric scoring, and shared bias across variants (relative deltas remain valid). The 50-example test set is small and bootstrap CIs are wide. The knowledge base is intentionally small (48 snippets); scaling it without redesigning the retriever would worsen context pollution, not improve it. **Safety statement.** This is a research prototype, not a clinical tool, and any generated report requires review by a licensed radiologist before clinical use.

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Ethics and Reproducibility

This is a research prototype, not a clinical tool, and any generated report requires review by a licensed radiologist before any clinical use. We use only publicly released datasets (NIH ChestX-ray14 [5] and Indiana University CXR [6]) under their original licences. No protected health information is committed to the public code repository. All generation runs locally so no patient data leaves the machine. All hyperparameters, prompts, and evaluation code are in the linked GitHub repository. The classifier trained on a single NVIDIA A100 in ~ 45 minutes; the full ablation evaluation ($50 \text{ examples} \times 3 \text{ variants}$) runs in ~ 15 minutes; the LLM-as-judge pass adds ~ 10 minutes.

LLM Usage Disclosure

The vision-language model used for report generation and as judge is Qwen2.5-7B-Instruct, served locally via the HuggingFace `transformers` library. During code development we additionally used Claude (Anthropic) and ChatGPT (OpenAI) for debugging assistance, code review, and clarifying questions about library APIs. All experimental design, dataset choices, evaluation methodology, and the contents of this report were authored by the team; LLM assistance was limited to code-level help.

Team Contributions

Ramya Chandrasekhara: dataset preparation (NIH), retriever design and implementation, evaluation harness, llm as a judge implementation, report writing.

Tyler Rutley: Data analysis (NIH), Indiana University Data Preprocessing, Multi modal Fusion Model (IU)

Faisal Mohammed Faisal Khan: VIT Multi-Classifer